



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology



Mechanics of
Composites
for Energy
and Mobility

Information
Science Lab

Qualcomm

Intelligent Structural Health Monitoring (SHM) using Semantic Edge AI and Mobile Sensing on Qualcomm Platforms

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PIMRC 2025 Demo

Motivation

High Energy Consumption

Continuous or periodic high-volume sampling drains battery life quickly, making long-term deployments difficult.

IoT Network Congestion

Transmitting raw, high-frequency data from many IoT devices overloads the communication network.



Heavy Edge Processing & Storage

Large amounts of raw sensor data require significant computational and storage resources at the edge node.

No Context-Aware Sensing

Basic IoT sensors cannot perform environmental or network-level analysis to decide when to collect high-quality data; they rely on fixed schedules instead of event-driven activation.

- Goal: Power-efficient, scalable system delivering timely,

Proposed Framework



Edge Risk-Based Control

Base Station or Central Server aggregates UAV findings, runs risk assessment, and selectively activates nearby IoT sensors.

03



AI-Powered UAV/Robot Inspection

Qualcomm RB5 on UAV performs real-time semantic segmentation to detect cracks, corrosion, leaks, vegetation, ruptures, ice/snow, and fire hazards.

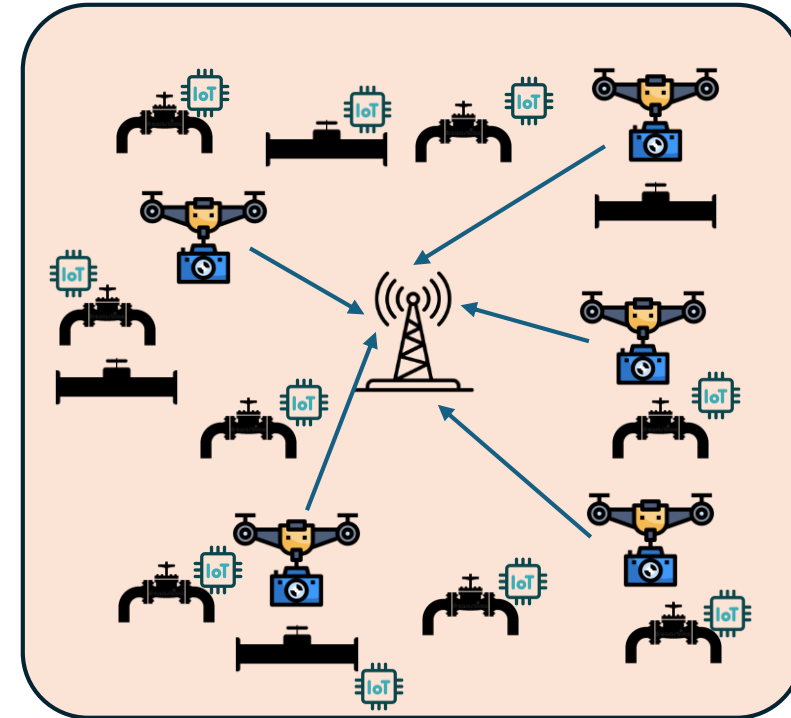
02



Event-Driven IoT Sensing

Ultra-low-power IoT nodes with Wake-up Radio, activated only when needed for on-ground measurements.

01



Our Approach

- ❑ Camera on drone first detects a crack using semantic segmentation (edge computing)
- ❑ Then a signal is sent to activate the sensor
- ❑ Sensor collects data and analyzed to confirm anomaly



RAPID VISUAL
INSPECTION



EVENT-DRIVEN
SENSING



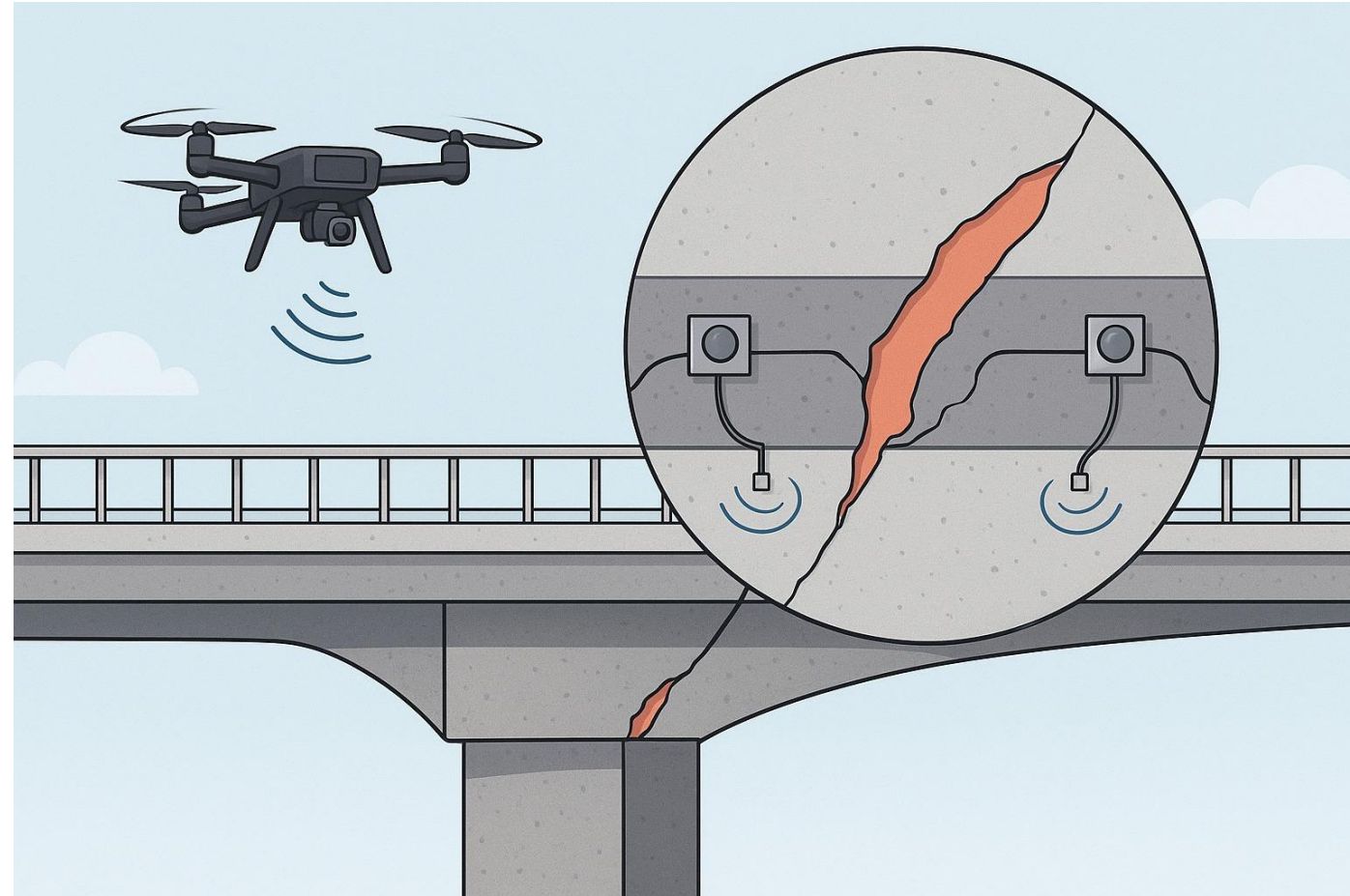
LOWER
MAINTENANCE
COSTS



SCALABLE TO
REMOTE/OFFSHORE
SITES

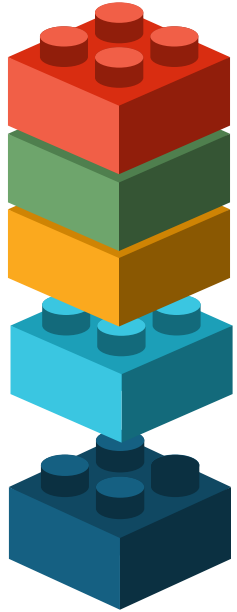


EARLY FAULT
DETECTION



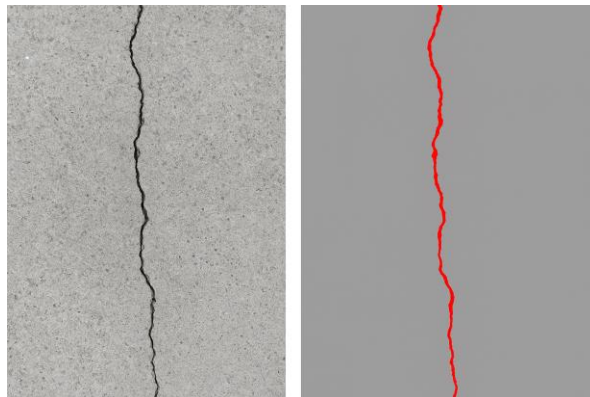
Semantic Segmentation for Crack Detection

Why we use it for crack detection?

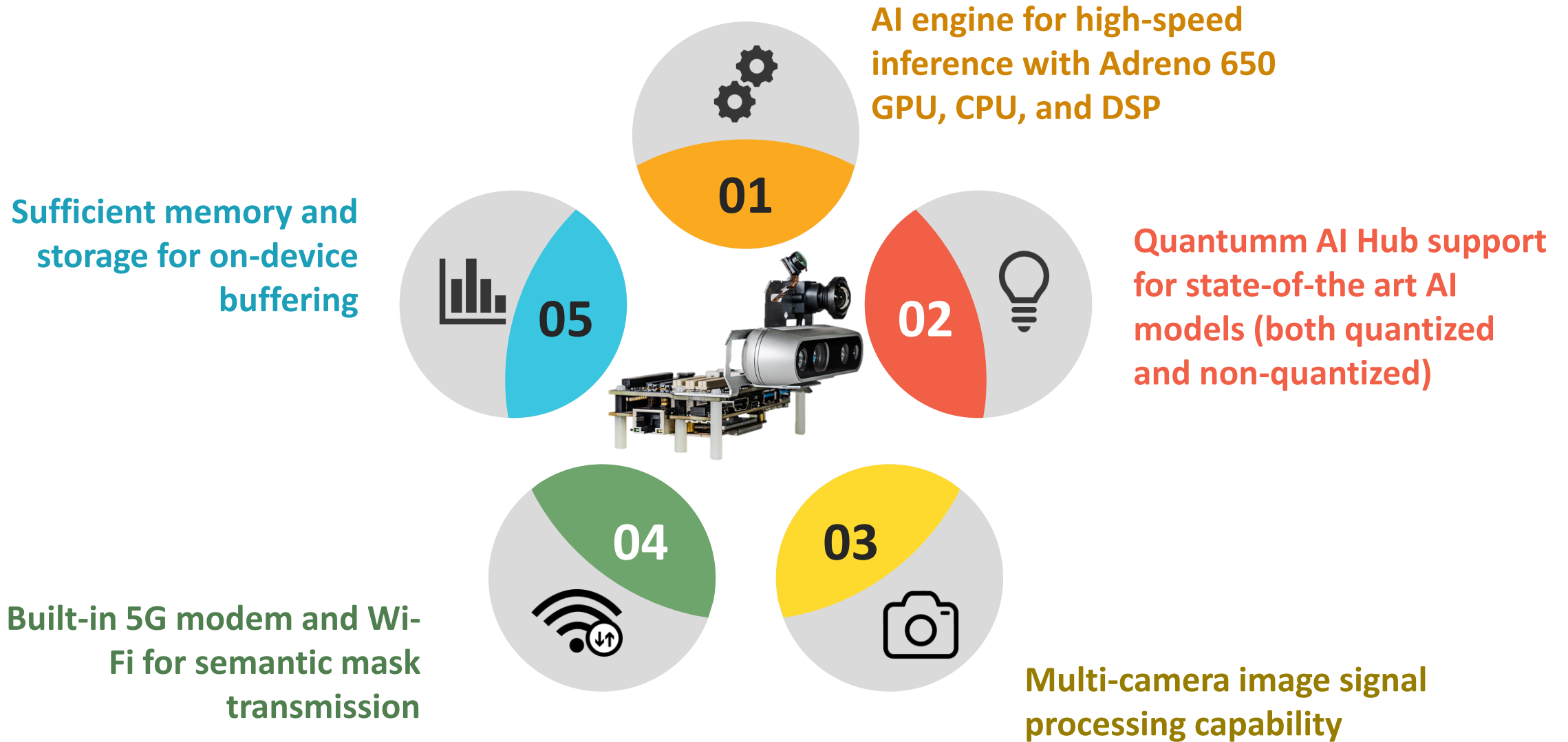


- **Pixel-level precision** → Detects fine, irregular crack shapes
- **Quantitative analysis** → Enables measurement of crack length, width, and area over time
- **Works on Complex Surfaces** → Handles varying backgrounds, lighting, and textures of infrastructure
- **Integrates with digital twin** → Segmented masks can be directly used for condition monitoring and predictive maintenance
- **Efficient transmission** → Only the segmented mask (small file) is sent instead of full-resolution images, saving bandwidth

Examples:



Why RB5?

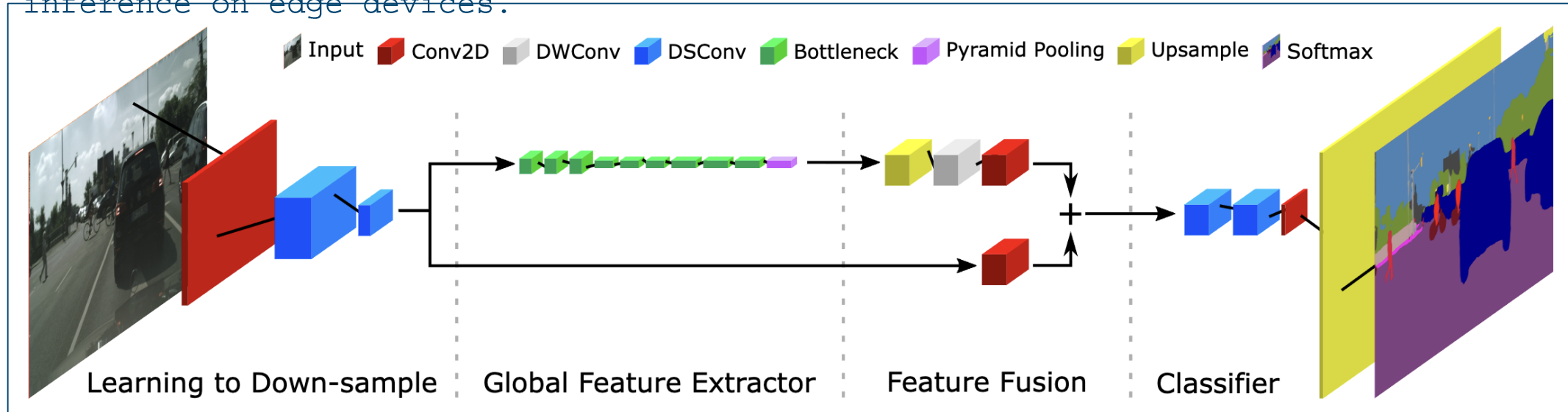


Semantic Segmentation Models

Model	Inference Time	Throughput (inferences / s)	Layers	Applicable Scenarios	Key Details
Segformer-Base	119 ms	8.42	544 NPU	Factory Automation Robotic Navigation	Transformer-based model
DeepLabXception	16.9 ms	59.3	207 NPU	Anomaly Detection Inventory Management	ResNet-101 and Xception backbones
DDRNet23-Slim	41.6 ms	24.1	133 NPU	Self-driving cars	Road-based scenes
EfficientViT-l2-seg	2.83 ms	0.35	462 NPU	Autonomous Driving Automotive	Transformer-based model
SAM 2	232 ms	4.32	647 NPU	Factory Automation Robotic Navigation	Zero-shot generalization
			⋮		
Many other on aihub.qualcomm.com					

Semantic Segmentation Model

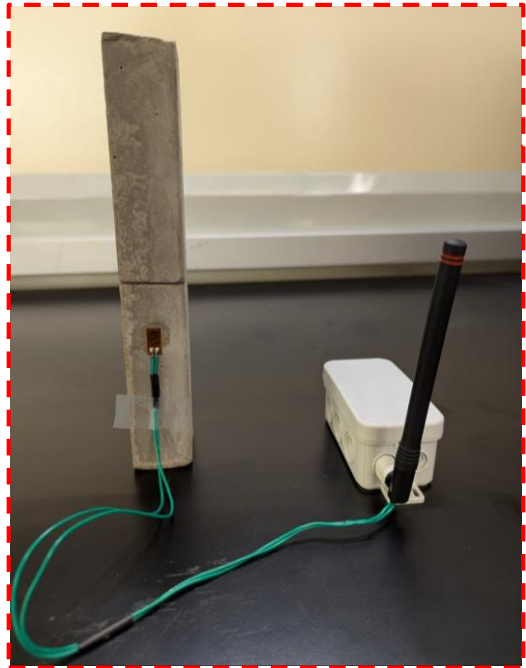
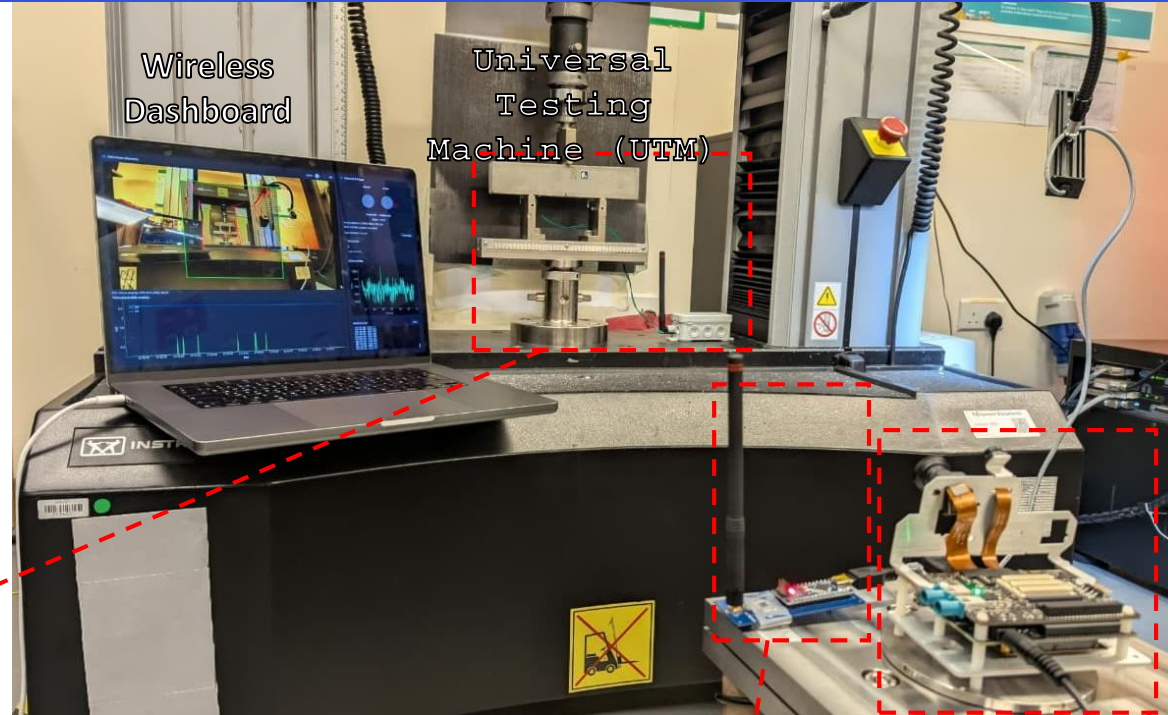
Fast-SCNN is **lightweight semantic segmentation network** designed for real-time inference on edge devices.



Fast-SCNN shares the computations between two branches (encoder) to build a above real-time semantic segmentation network.

- **Real-time Performance** → Processes high-resolution images at edge in milliseconds.
- **Low Computational Cost** → Small parameter size enables deployment on embedded AI accelerators.
- **Fine-Grained Detection** → Captures thin, irregular crack patterns at pixel level.

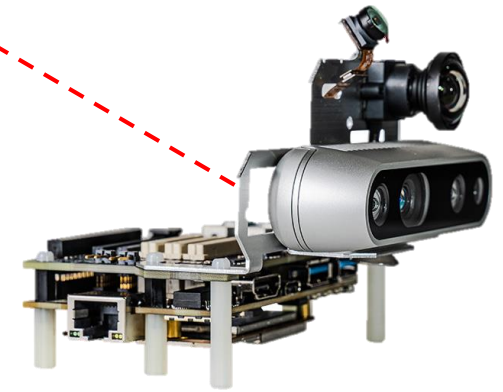
System Components



Sensor Layer: Strain, vibration, acoustic, temperature, humidity sensors + wake-up receiver

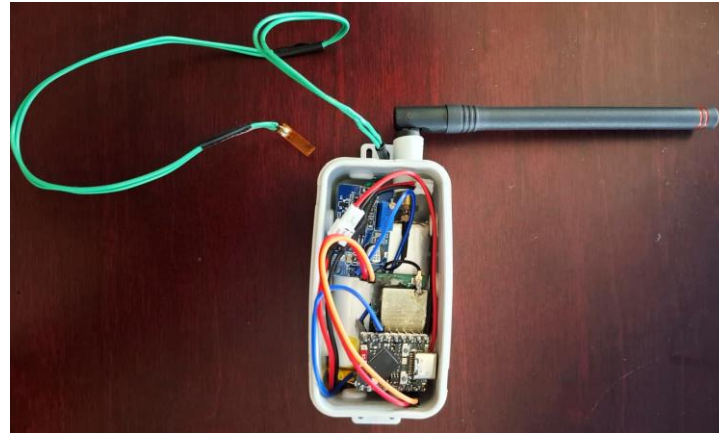


Wu-Tx: Wake-up transmitter for enabling event driven sensing

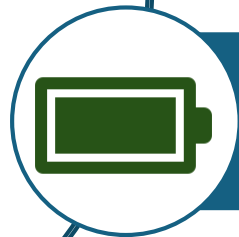


Edge Layer: RB5 for semantic crack segmentation, mobile inspection, and event driven sensing

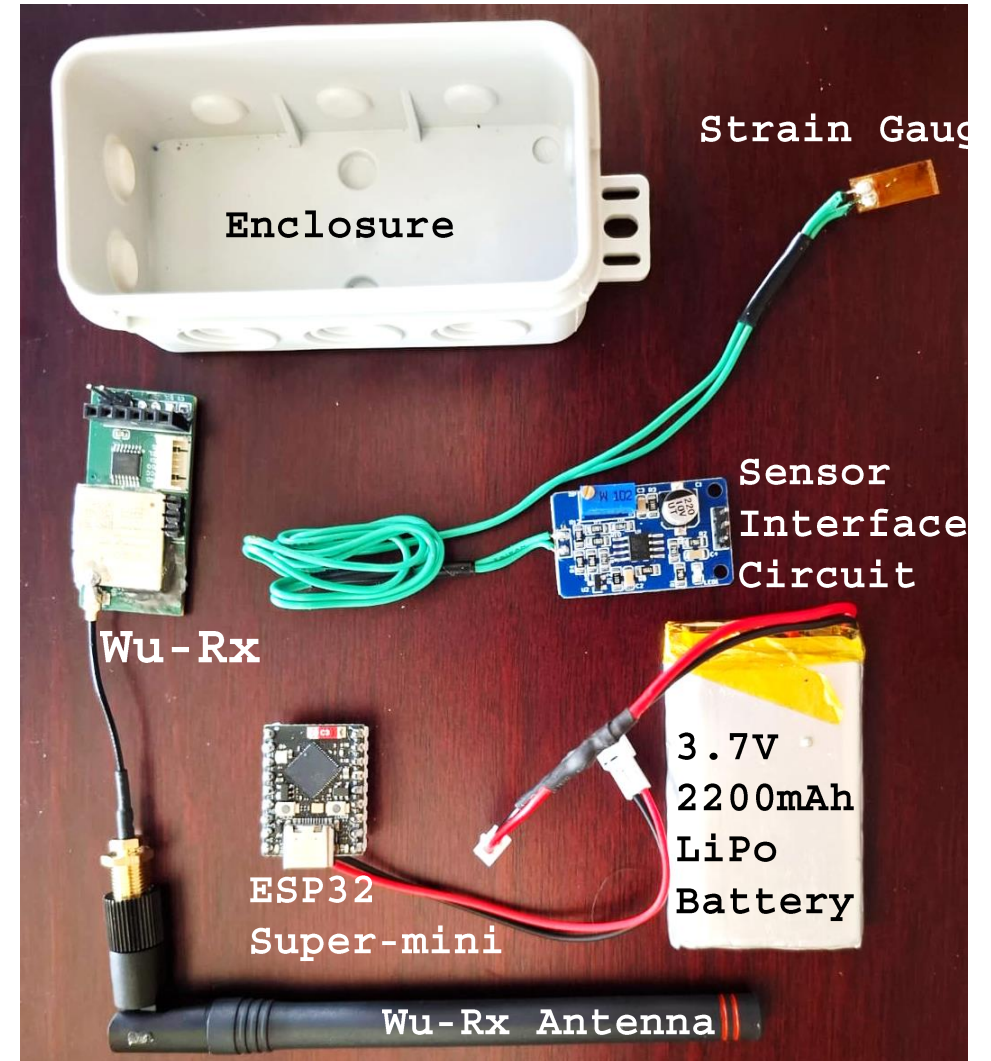
IoT Components & Power Consumption



Total Deep Sleep
Power
Consumption: 50
µA



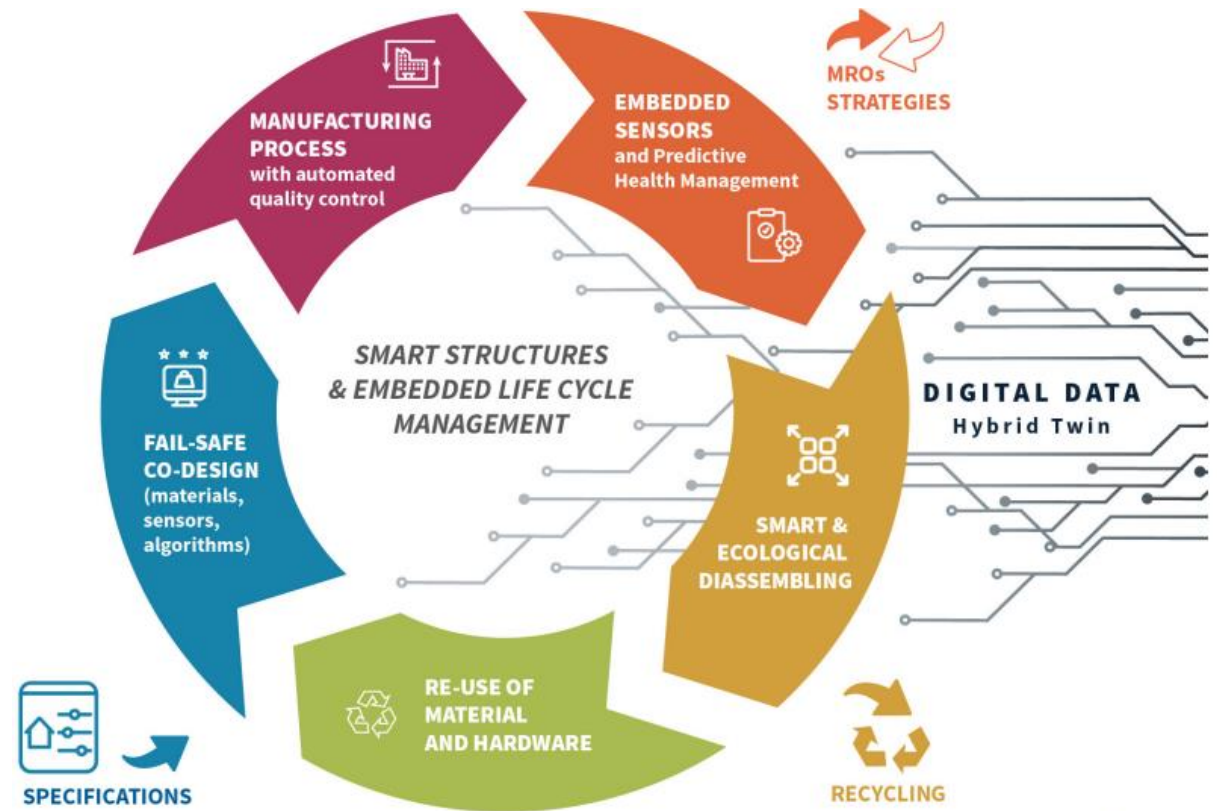
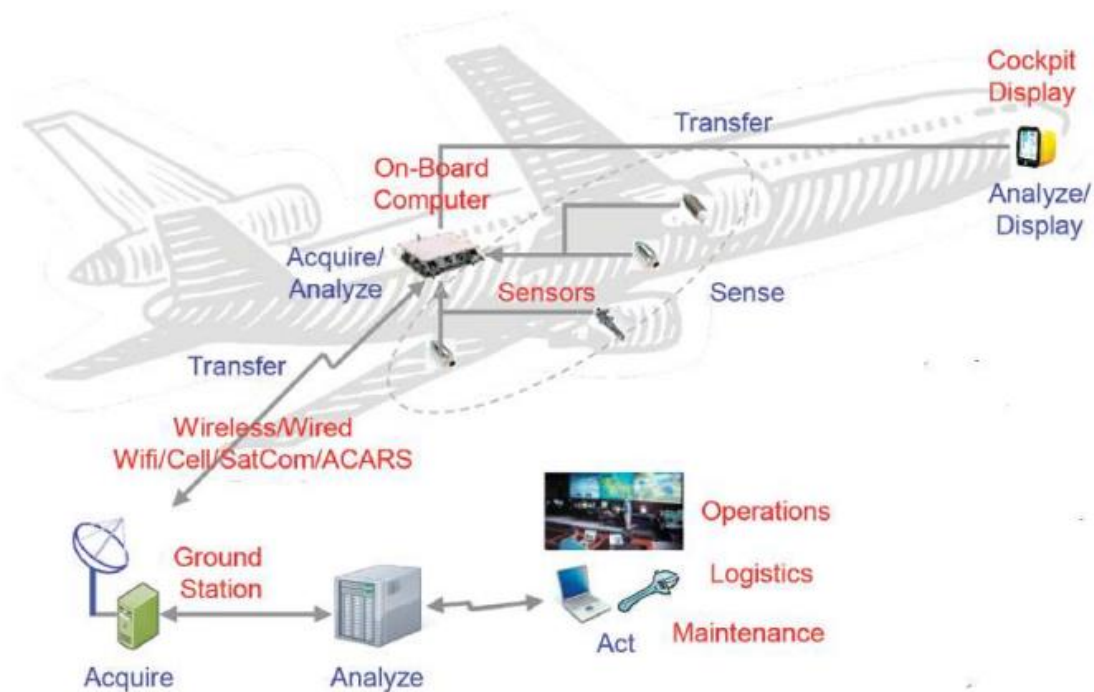
Deep Sleep
Lifetime (2200mAh
Bat.): 5 Years



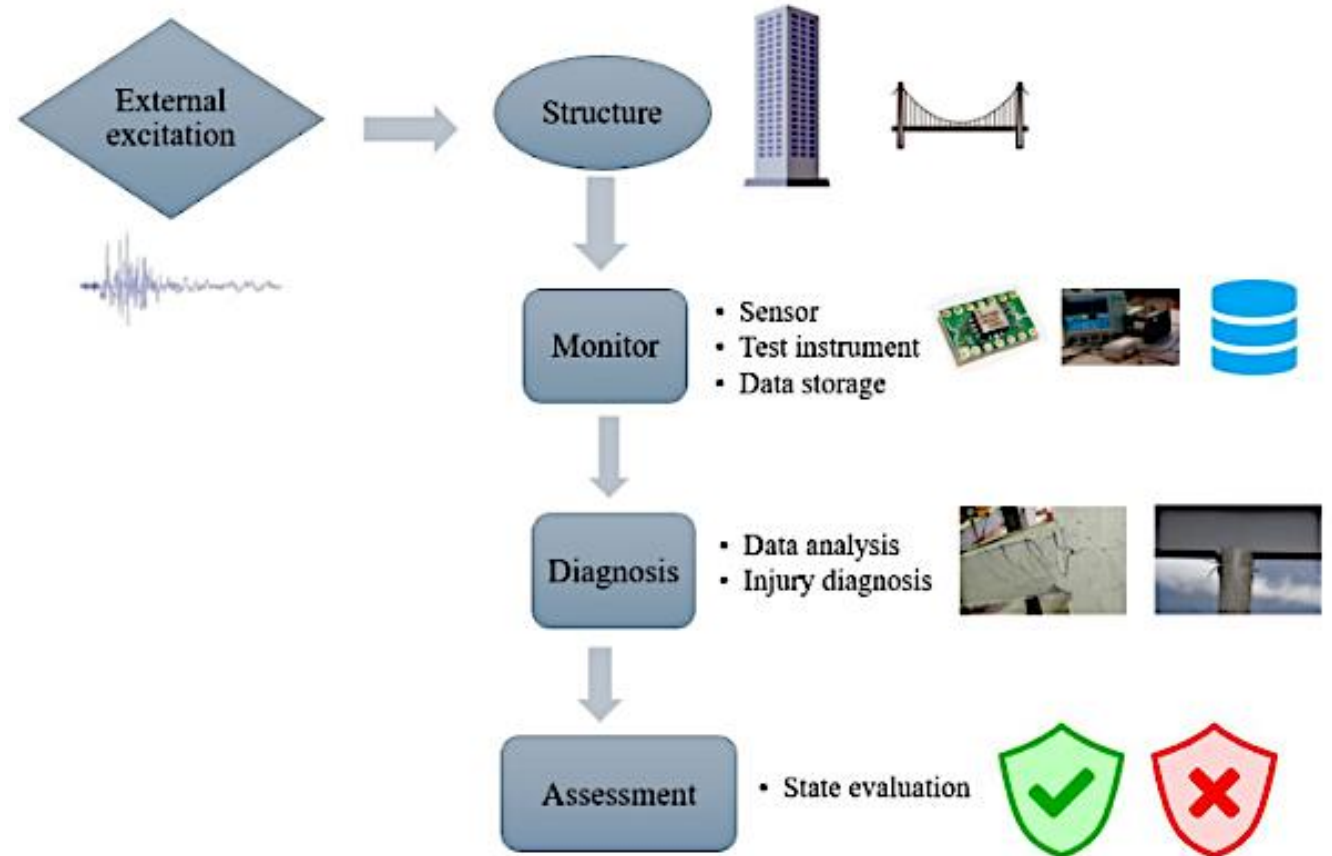
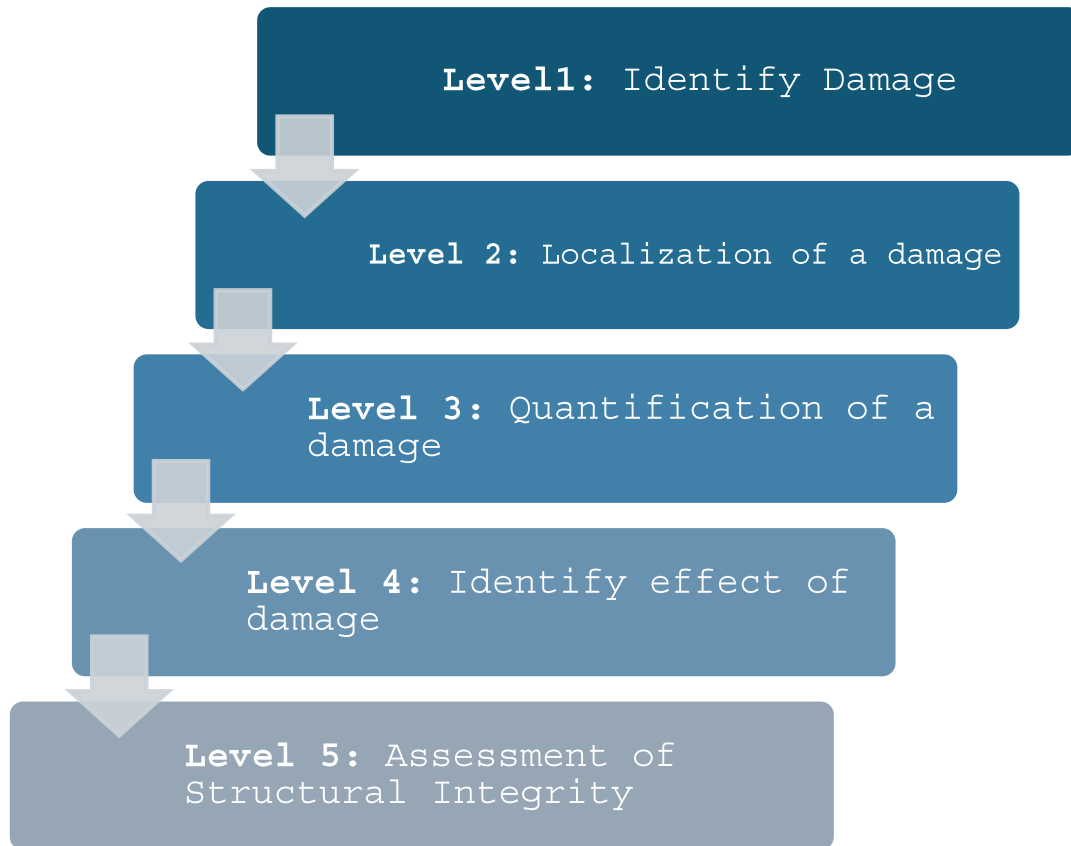
SHM Concept: From Definition to Objectives

"The process of acquiring and analyzing data from on-board sensors to evaluate the health of a structure" - "Guidelines for Implementation of Structural Health Monitoring on Fixed Wing Aircraft", ARP6461-B, published by SAE International (August 2021)

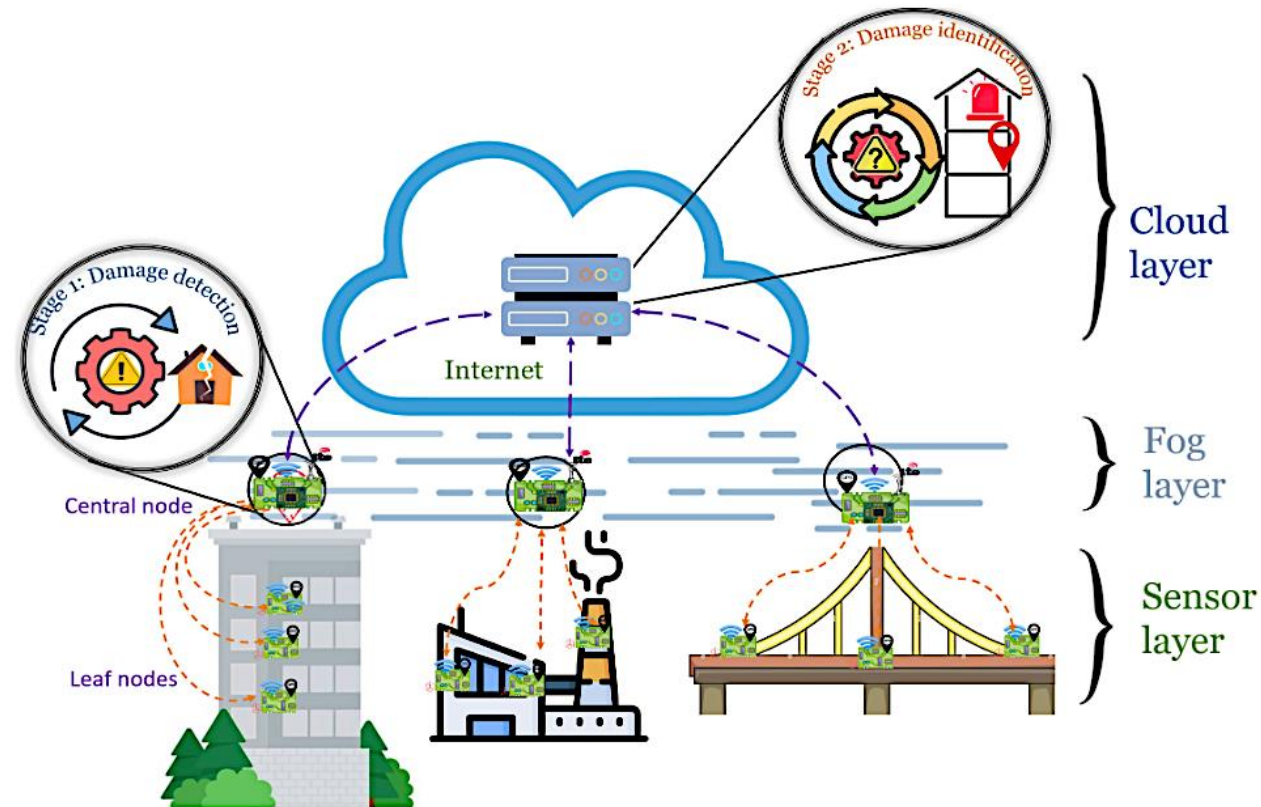
Onboard and off-board elements of IHM



Damage Detection Levels of SHM



Edge Computing for SHM Applications



Real-Time
Local
Processing



Intelligent
Data
Filtering



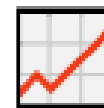
Enhanced
Power
Efficiency



Improved
Reliability



AI/ML at
the Edge



Scalable
Architectur
e



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Thank you!